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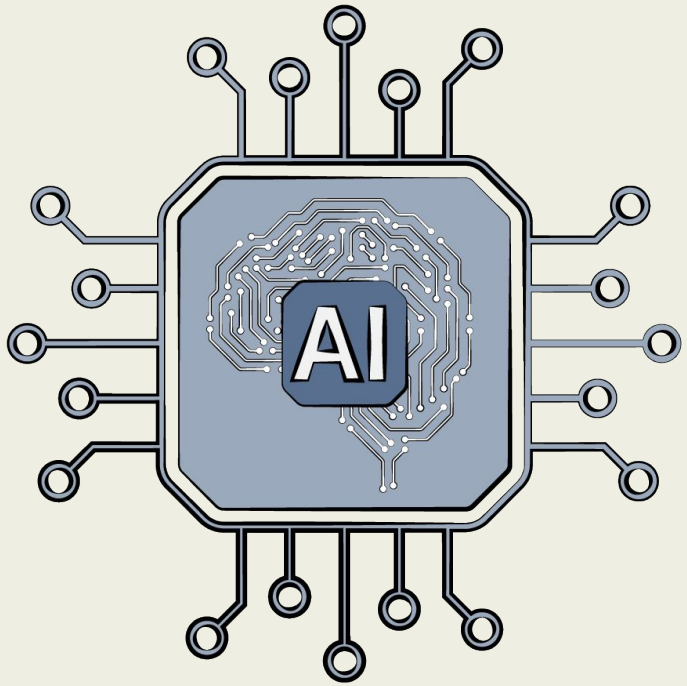
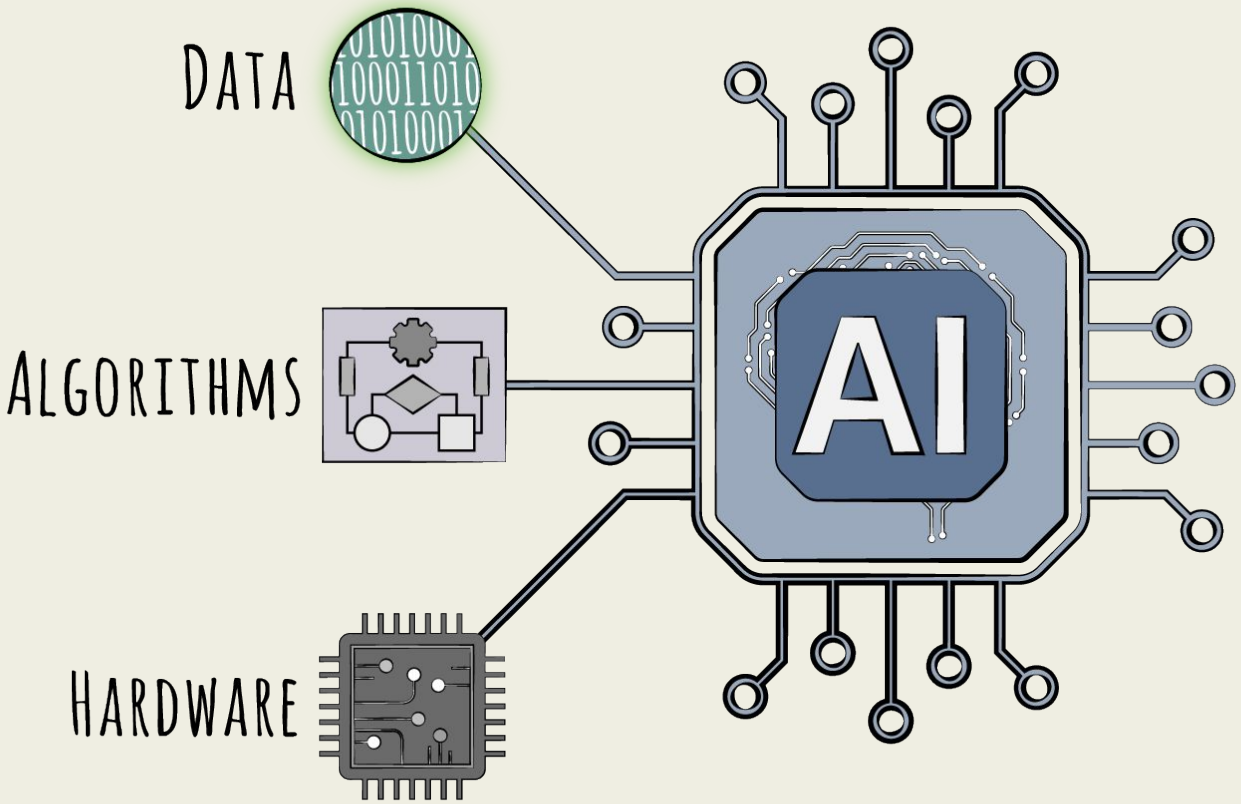
Artificial Intelligence (AI) Primer Summary Slides



Introduction



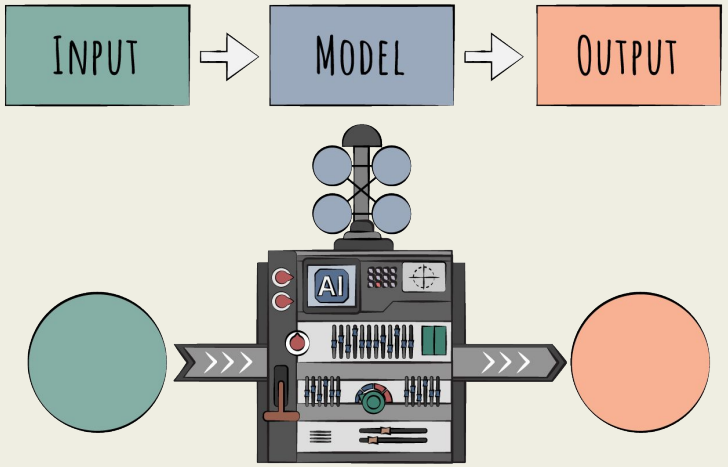
Three Drivers of AI Innovation

1. **Data Proliferation**: Vast increases in available digital data (images, text, etc.)

2. **Algorithm Advancement**: Enhanced computing algorithms that leverage this data to train AI models

3. **Computing Hardware Development**: High-powered machines capable of processing massive data sets quickly and efficiently (e.g., using advanced GPU semiconductors)

AI Model Fundamentals



Building → **Applying**

- Input variables
- Mathematical form
- Parameters

AI Model Fundamentals

What is an AI model?

- A model simplifies reality to make decisions or predict outcomes
  - Examples: financial models for budgeting, structural models for construction
- AI models means one that mimics aspects of human intelligence, operating as input-output machines

How do AI models work?

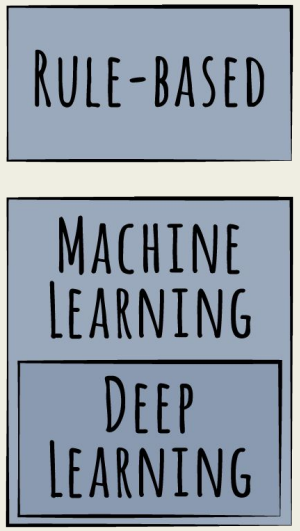
- 2 steps: build the model, then apply it
- Take an example: predicting sales for a new retail store
- Build:
  - Input variables: select key factors (e.g., store size) influencing the output
  - Mathematical form: Define how inputs are converted to outputs (e.g., linear relationship)
  - Parameter optimization: Determine the best values for model parameters (e.g., using regression) to fit the training data
- Apply: use the model to estimate outcomes based on new inputs

How can you improve AI models?

- Enhance prediction accuracy:
  - Adjust parameter values through re-training on larger datasets
  - Introduce more input variables and other mathematical forms to capture complex patterns
- Avoid overfitting:
  - Overfitting occurs when a model fits a smaller training dataset too closely, failing on new data
    - More parameters generally require more training data to avoid overfitting

AI Domains

- The AI world of AI is vast and can be segmented into various subfields or domains, often overlapping and defined inconsistently
- Ways to categories AI domains: by input types, by output tasks, or by model types used
  - By input, e.g.:
    - Natural Language Processing (NLP): focuses on human language inputs, crucial for applications like voice recognition and translation.
  - By output, e.g.:
    - Generative AI (GenAI): Specializes in creating new content (text, image, audio), with examples including ChatGPT
  - By model type:
    - Rule-based AI: Early AI models using hardcoded if-then logic. Humans encode the rules, and the model executes.
    - Machine Learning (ML): Includes models that learn from data to determine the best rules or patterns
      - Model Development Steps in ML:
        - Input variable selection and mathematical form definition by humans.
        - Parameter optimization via machine training
      - Types of ML Models:
        - Ranges from simple forms (e.g., linear regression) to complex structures (e.g., neural networks).
          - i. Neural Networks: model architecture that mimics the human brain structure, processing through layers of nodes or neurons
        - Deep Learning is a subset of ML: models that use larger neural networks
          - i. Focused on more complex problems requiring advanced model architectures
          - ii. Predominantly used in cutting-edge AI applications (e.g., LLMs, image-generating models)



Training Approaches

Training Approaches

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning (RL)

The "intelligence" of machine learning models depends on their ability to learn from training data; training involves optimizing parameters to best fit the training data

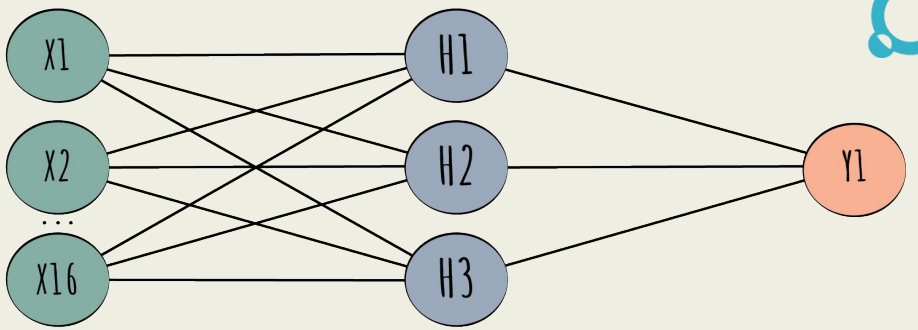
Types of training approaches:

- Supervised Learning
  - Utilizes labeled data (where each input is paired with a known output)
  - Models learn to match their outputs with the correct answers, refining parameters during training
  - Example: Image recognition models determine the presence of objects (like the letter 'T') in images using a labeled training set
- Unsupervised Learning
  - Operates on unlabeled data where outputs are not predefined
  - Useful when labeling data is impractical, or the nature of the problem does not permit predefined outputs
  - Example: customer segmentation models group profiles based on detected patterns without prior output labels
- Reinforcement Learning (RL)
  - Trains model through feedback rather than from direct data labels
  - Can be used in situations where traditional training data is insufficient or ongoing adaptation is required
  - Example: AlphaGo's training involved rewarding winning strategies and penalizing losses
  - Self-driving cars use RL by receiving rewards or penalties based on maneuver success
    - Reinforcement Learning from Human Feedback (RLHF): A form of RL where models adjust based on direct human feedback

Many models use multiple training approaches, e.g. LLMs, which are trained in two steps:

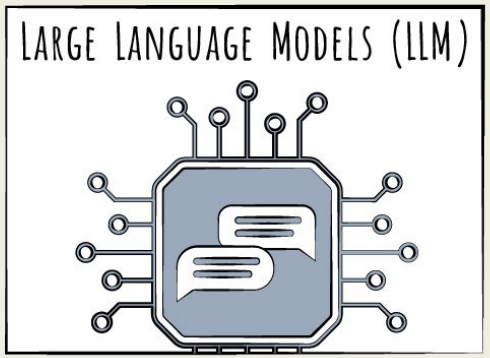
- Pre-training
  - Involves learning from large volumes of unlabeled text to recognize and replicate language patterns. The 'P' in GPT stands for pre-trained
- Fine-tuning
  - Uses supervised learning to specialize the model on tasks like sentiment analysis by training on labeled examples
  - Often incorporates RLHF, enhancing model responses based on user feedback

Understanding Neural Networks



- Neural Networks: a type of AI model that mimics the configuration of the human brain, moving information across layers of neurons from the input to the output; critical in advanced AI models like image generators and language processors
- How to build a Neural Network (using an example: an image-recognition model that detects the letter T)
  - Input data: Images that either do or do not contain the letter T
    - Images are unstructured data and must be “structured” (transformed into a long single row of data)
      - e.g., converting a black and white 4x4 pixel image into a structured format means transforming it into a single row with 16 columns (1 for black, 0 for white)
  - Neural Network design (continuing with the example)
    - Layers:
      - Input layer: 16 neurons (X1 to X16).
      - Hidden layer: One hidden layer with three neurons (H1, H2, H3) (this is a design choice, our example is simple)
      - Output layer: In our example, a single output neuron (Y1) indicates if the image is a 'T' (output as probability)
    - Neuron calculation: What are the values of each neuron? This is a design choice (the function used is called the activation function). In our example, we use the sum of the previous layer’s outputs, adjusted by weights (which are the parameters).
  - Training the model
    - Process: Adjust weights based on training data (supervised learning) to maximize accuracy
      - Usually, training begins with an initial guess and refines the parameters iteratively until the accuracy of the model improves.
- Real-world applications: Much larger than our sample
  - Layer image inputs (color images lead to high-dimensional data (up to millions of inputs)
  - This requires larger neural networks with more layers (and therefore more parameters)
  - Deep Learning: when Neural Networks with more than two hidden layers are used
    - Complex NN architectures like Convolutional Neural Networks (CNNs) for images and other specific architectures for text-based models

Understanding LLMs



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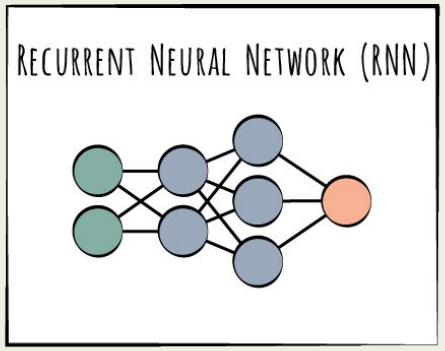

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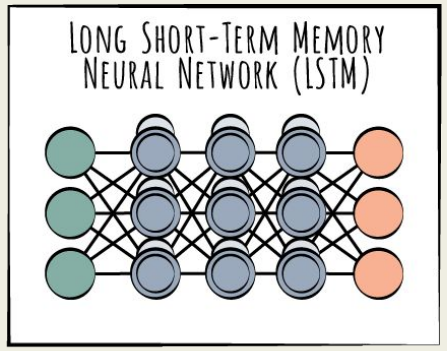

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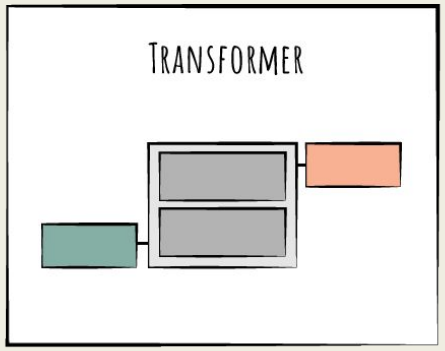

Understanding LLMs

- Images and text require different types of Neural Network (NN) architecture
- How do most text-based NN models work?
  - Outputs are generated by finding the most likely word to follow in a sentence or phrase, based on the recent context
    - Generating this output requires thousands of different output-layer neurons, one for each common possible word
  - The single most likely word can be chosen every time, or there can be some randomness (called 'temperature') in word selection; temperature enhances the quality of generated content, making it less predictable
  - In practice, models do not work one word at a time, but one token at a time
    - A token is a text fragment that can represent anything from a punctuation mark or part of a word to a full word
  - In any case, word sequence is key (each word depends on its predecessor), and this is part of what makes LLMs unique from image models
- What common NN approaches make Large Language Models (LLMs) possible? There are 3 common NN architectures used for text
  - Recurrent Neural Networks (RNNs):
    - Designed to handle sequence data by maintaining a form of memory across inputs
    - Challenges include “memory” of the context fading with long sequences and limited ability to work via parallel processing
  - Long Short-Term Memory Networks (LSTMs):
    - An advanced type of RNN that includes mechanisms to manage “memory” more effectively, using components like the “forget gate”
    - Still faces challenges with parallel processing
  - Transformers:
    - Introduced in 2017, addressing both memory retention and scalability (can be parallelized)
    - Utilizes “attention” mechanism to focus on relevant parts of input data, enhancing processing efficiency
    - Dominant architecture in modern LLMs due to its suitability for handling lengthy text sequences; GPT is a transformer (that’s the T in GPT)
- Since transformer models are parallelizable, they are suited for use with parallel processing chips like GPUs

Trends

New Deep Learning models

- Expansion of Model Capabilities:
  - Increasing size and complexity, requiring more computational resources
  - Improving in handling longer contexts and performing multi-tasking, including zero-shot learning
- Evaluation and Standardization:
  - Use of standardized tests like MMLU to assess model capabilities
- Diverse Model Developments:
  - Shift towards smaller, specialized models (SLMs) for cost-effective, targeted applications
  - Development of multi-modal models that handle both text and visual data, like OpenAI's SORA and Google's Gemini

New applications of Deep Learning models

- Chain-of-Thought and AI Agents:
  - Enhanced task execution through sequential prompting and integration with other software via APIs
  - Development of AI agents capable of managing complex tasks like calendar organization and priority inbox sorting
- AI in research and discovery
  - AI's role in accelerating research, notably in drug discovery with models like AlphaFold predicting protein structures
  - Potential to significantly speed up breakthroughs in other fields such as fusion power or new materials

AI safety (Responsible AI)

- Accuracy and reliability:
  - Efforts to mitigate AI 'hallucinations' through techniques like Retrieval-Augmented Generation (RAG)
- Bias and ethical concerns:
  - Addressing inherent biases in AI models and exploring frameworks like Constitutional AI for principled decision-making
- Intellectual Property and misuse:
  - Legal and ethical challenges regarding the use of proprietary data and the potential for AI misuse in creating misleading information

Glossary I

Activation Function: A mathematical function used in neural networks to calculate the output of each neuron from its input data
AI Agents: Advanced AI applications that automate and manage tasks or workflows, often through integration with other digital tools
AI Model: A computer model that mimics human intelligence by generating machine outputs from given inputs
Chain-of-Thought: A method where an AI model is prompted sequentially to perform complex tasks by building on previous responses
Constitutional AI: An approach where AI behavior is guided by a set of underlying principles to ensure ethical decision-making and mitigate biases
Convolutional Neural Network (CNN): A type of neural network particularly effective for processing structured grid data like images, using layers that automatically and adaptively learn spatial hierarchies of features
Deep Learning: An advanced branch of machine learning that uses deep neural networks to handle complex tasks
Diffusion Models: Advanced neural network architectures used for generating high-quality and coherent images or videos by learning the distribution of training data and iteratively refining generated outputs
Fine-tuning: A subsequent phase of model training using targeted data to refine capabilities on specific tasks or to improve performance on detailed aspects
Generative AI (GenAI): A branch of AI focused on generating new digital content from existing data
High-dimensional Data: Data represented by a large number of attributes or dimensions, often derived from unstructured sources like images
Input Variables: Factors considered by a model to influence its outputs, such as store size in sales predictions
Large Language Models (LLMs): A type of deep learning model specifically designed to process and generate human language
Long Short-Term Memory (LSTM): An RNN variant that includes mechanisms to remember and forget information selectively, aiding in handling longer sequence
Machine Learning (ML): AI models that learn from data to improve their accuracy without being explicitly programmed for every scenario
Mathematical Form: The mathematical equation or function defining how inputs are transformed into outputs
Multi-Modal Models: AI models capable of processing and understanding multiple types of data inputs, such as text and images
Natural Language Processing (NLP): AI domain dealing with the processing and understanding of human language
Neural Network: A network of nodes (or artificial neurons) that process data in layers, emulating the human brain's structure

Glossary II

Overfitting: A modeling error which occurs when a model is excessively complex, fitting the training data too closely but performing poorly on unseen data
Parameters: Values within a model that are optimized during training to best fit the data
Pre-training: The initial phase in training a model where it learns from a broad data set without specific targets to develop a general understanding
Recurrent Neural Network (RNN): A type of neural network that processes sequences by maintaining a state or memory of previous inputs
Regression: A statistical method used to fit models to data, commonly used to find optimal parameter values
Reinforcement Learning (RL): A training strategy where models learn through trial and error, receiving rewards or penalties based on their performance
Reinforcement Learning from Human Feedback (RLHF): A variant of RL where human feedback directly influences the training process, guiding the model's learning
Retrieval-Augmented Generation (RAG): A technique where AI models enhance their responses by cross-referencing with up-to-date external data sources to improve accuracy
Rule-Based AI: AI models that operate on predefined rules set by developers
Small Language Models (SLMs): Smaller, more efficient models designed for specific tasks, requiring less computational power than larger models
Supervised Learning: A machine learning approach where the model is trained on a dataset containing inputs paired with correct outputs
Temperature: A factor in LLMs that introduces randomness into the decision-making process, affecting the selection of output tokens
Token: The smallest unit of processing in many LLMs, varying from parts of a word to entire words
Training Set: The dataset used to train a model, allowing it to learn from known input-output pairs
Transformer: A neural network architecture that uses attention mechanisms to dynamically focus on different parts of the input data, suitable for large-scale and complex tasks like those needed in LLMs
Unsupervised Learning: Training method using datasets without predefined labels, allowing the model to identify patterns or structures independently
Zero-Shot Learning: Ability of a model to perform tasks it has not been explicitly trained to do